CS & IT ENGINEERING

Operating System

Basics of OS

ONE SHOT

Topics to be Covered

- **Topic** Operating System Definition
- **Topic** Types of Operating System
- **Topic** Dual Mode of Operation

Topic : Introduction

- **GATE Ranks:**
  - 682 (2009) – 3rd year
  - 19 (2010) – 4th year
  - 119, 440 etc.

- **Education:**
  - ME from IISc Bangalore
  - M. tech from BITS-pilani in Data Science

- **Work:**
  - 18+ Year Teaching Experience
  - 15+ in GATE/IES
  - Worked in Cisco, Audience Communication

Daily lectures ⇒ Notes ⇒ Daily revision

saturday – weekly revision

sunday – weekly test

chapter over – PYQs Practice

6-8 M

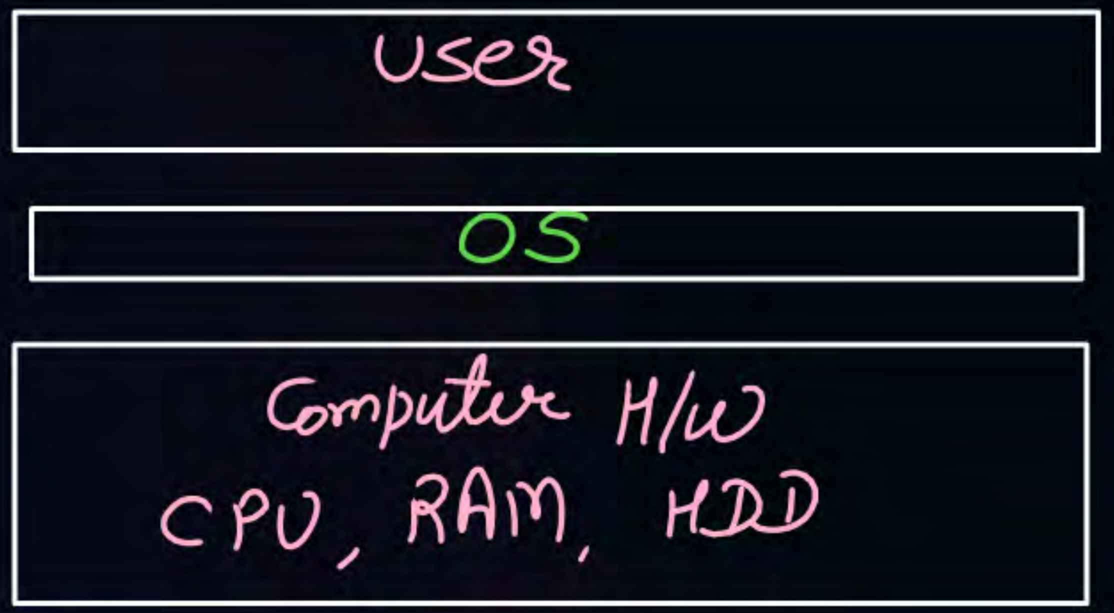
Prerequisite ⇒ Number system

- → Binary
- → Hexadecimal

COA ?

Topic : Operating System

Interface between user and hardware



Topic : Operating System

- Set of utilities to simplify application development/execution

→ API

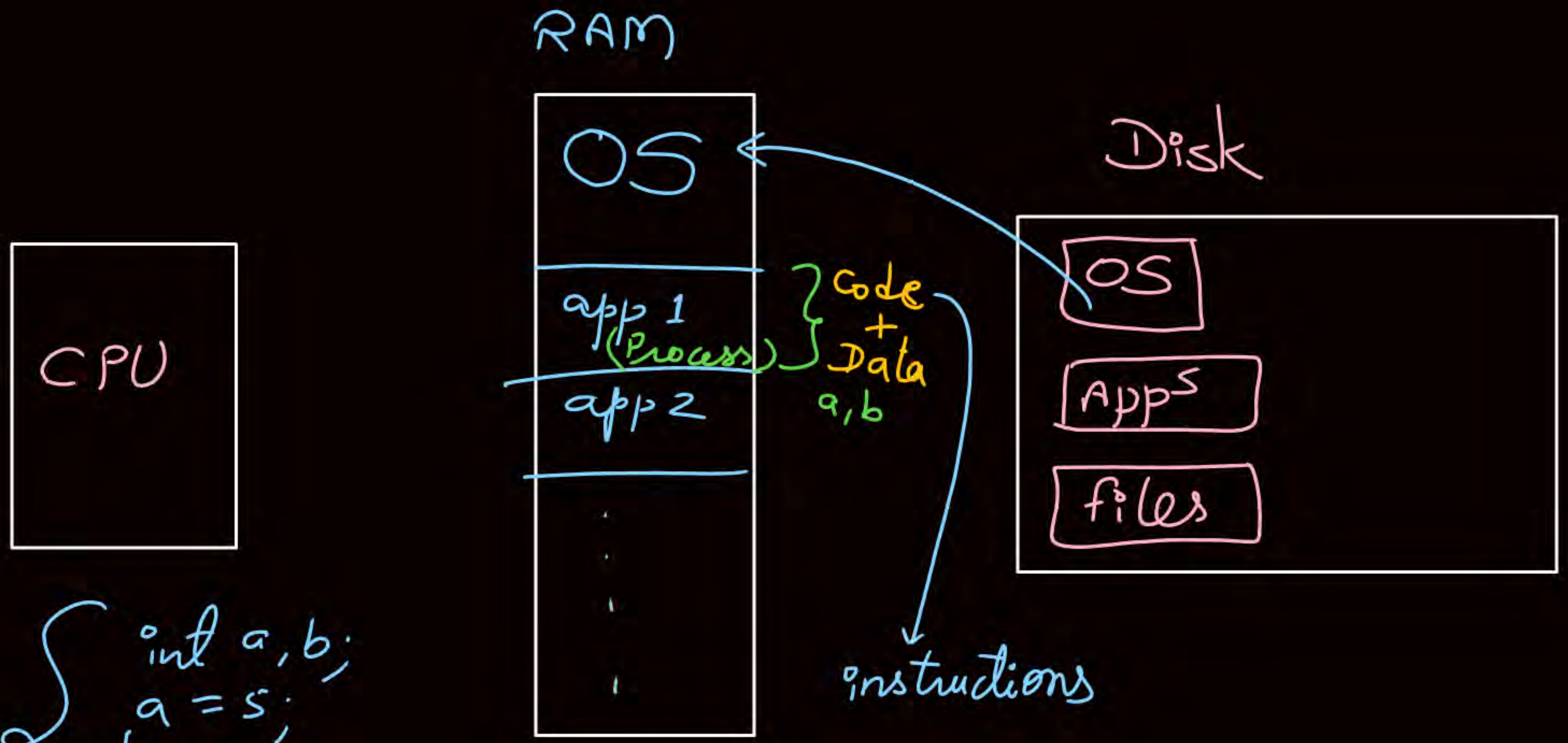
CPU
RAM
I/O device ⇒

Topic : OS Course Syllabus

Chapter Number	Chapter Name
1	Introduction
2	Process Management
3	CPU Scheduling
4	Process Synchronization (IPC)
5	Deadlock
6	Memory Management & Virtual Memory
7	File System
8	Disk Scheduling

Topic : Services of OS

- User Interface
- Program Execution
- I/O Operation
- File-System Manipulation
- Communication (Inter-process Communication)
- Error Detection
- Resource Allocation → file, device, memory space
- Accounting
- Protection & Security




```
int a, b;
a = 5;
b = 10;
printf("%d", a+b);
```

Topic : Types of OS

1. Uniprogramming OS
2. Multiprogramming OS
3. Multitasking OS (Time Sharing)
4. Multiprocessing OS
5. Multiuser OS
6. Real Time OS
7. Embedded OS
8. Handheld Device OS

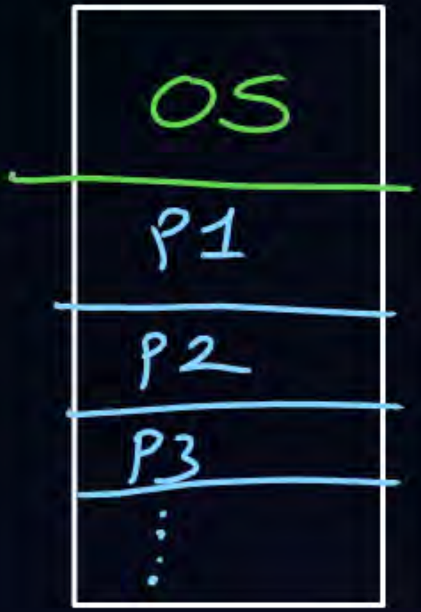
Topic : Uniprogramming OS

It allows only one program in main memory (RAM).

If the single program goes for I/O operation then CPU will be free (idle).

Topic : Multiprogramming OS

It allows more than one program to be in m.m. (RAM) at a time.



if one program goes for I/O operation then other program can use CPU.

Degree of multiprogramming :-

No. of programs in m.m. at a time.

As the degree of multiprogramming increases, CPU utilization also increases but upto a certain limit.

Types of multiprogramming OS

Non-preemptive

A running program leaves CPU only when it wishes to.

preemptive

A running program can be forcefully taken out of CPU against wish.

Topic : Multitasking OS

Extension of multiprogramming OS in which processes execute on CPU in time shared (round-robin) manner.

P1	P2	P3	P4	P5	P1	P2	P3	P4	P5

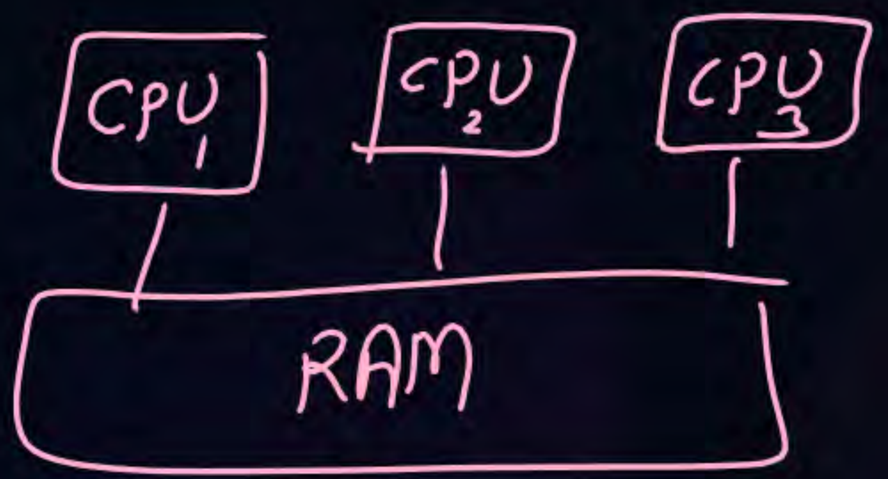
...

Topic : Multiprocessing OS

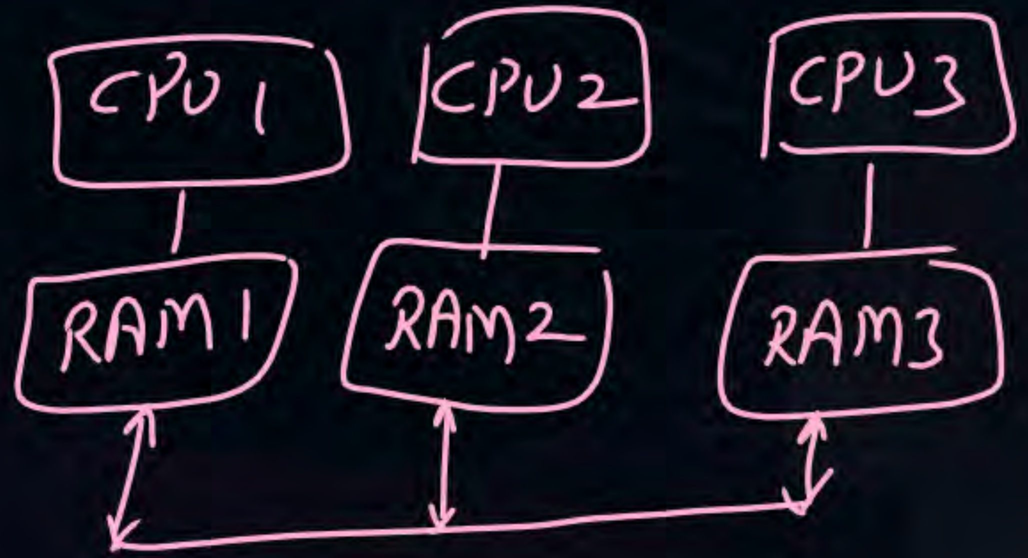
This type of OS works on computers which have multiple CPU.

Types

Tightly coupled
(shared memory)

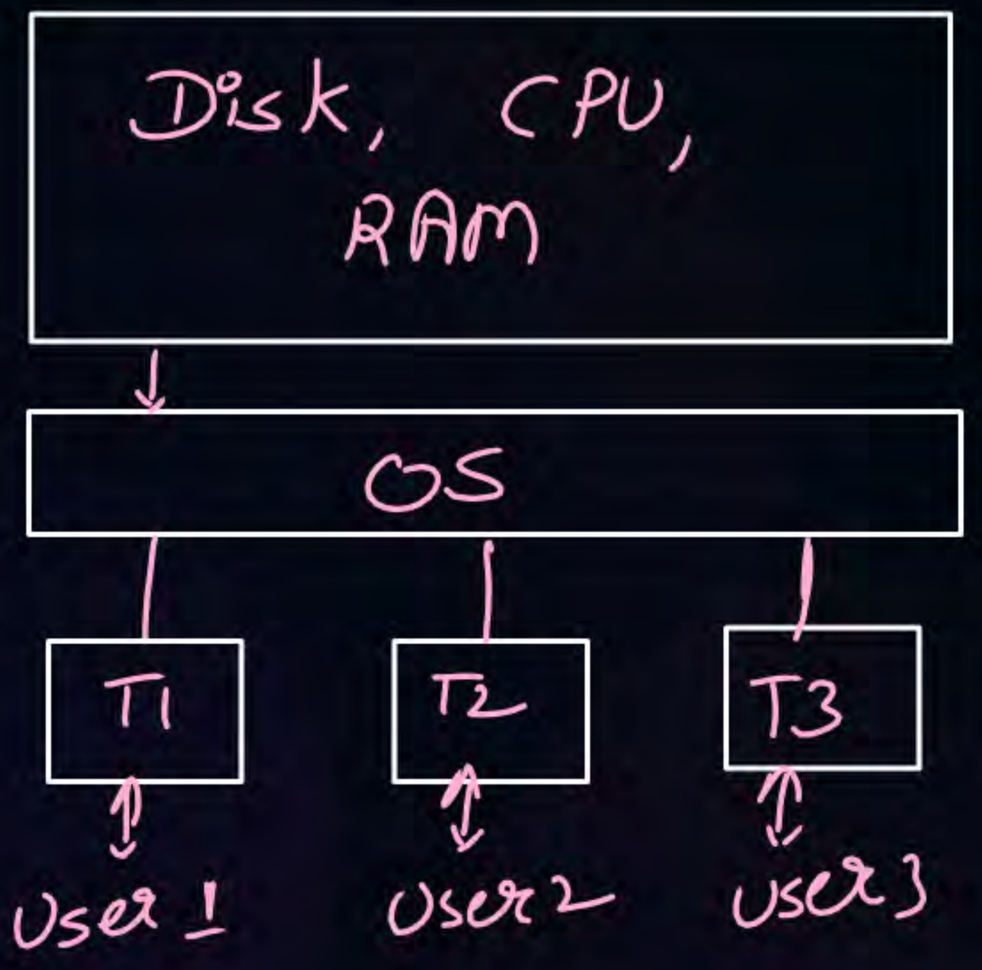


Loosely coupled
(distributed system)



Topic : Multiuser OS

This OS allows more than one user to use single computer at a time.



T1, T2, T3 ⇒ Terminal (screen, keyboard, mouse & I/O devices)

Topic : Real Time OS

works on such computer system which works on real time data or event.

OS provides deadline to each & every process.

Types

soft real time OS

deadline ⇒ lineant

Hard real time OS

deadline ⇒ strict

Topic : Embedded OS

OS provided with embedded OS.

Topic : Handheld Device OS

OS works on handheld devices.

2 mins Summary

Topic — Operating System Definition

Topic — Types of Operating System

Topic — Dual Mode of Operation

Happy Learning

THANK - YOU